



RAG with SAKURA AI Engine

SAKURA AI Engine Practical Exercise Module 1

<https://www.sakura.ad.jp/>

DATE

2026.9

COMPANY

SAKURA internet Inc.

SERVICE

SAKURA AI Engine

VERSION

Ver.1.0

Today's Agenda

SAKURA AI Engine Practical Exercise consists of four modules, 1 through 4.

Module 1

SAKURA AI Engine Practical Exercise Module 1

★ You are here

Log in, select a model, issue an Account Token, run inference, and build a RAG chatbot using a document.

Module 2

SAKURA AI Engine Practical Exercise Module 2

Converting audio to text (transcription): Comparing accuracy in English and Thai to summarize.

Module 3

SAKURA AI Engine Practical Exercise Module 3

Combine image and text input to design AI output that supports business decisions.

Module 4

SAKURA AI Engine Practical Exercise Module 4

Embed image and text AI calls into a web app and run it in a browser.

Goals for This Class

Go one step beyond asking AI questions. Build an AI system that answers using your own documents.

1 Log in to SAKURA AI Engine and select a model

Review the available models, choose one, and try it.

2 Issue an Account Token and run a request with curl

Experience calling AI from a program.

3 Register a document for RAG

Make your own document available to AI.

4 Explain the difference between using RAG and not using RAG

This is the most important goal of the class.

Completing these four goals brings you one step closer to building AI into applications.

Schedule (90 Minutes)

In this class, you will log in to SAKURA AI Engine and use commands to run an AI system that answers based on your own documents.

Time	Content
0 - 10	Overview 1: What Is SAKURA AI Engine?
10 - 25	Hands-On 1: Log In and Use Playground
25 - 35	Overview 2: What Is an API?
35 - 55	Hands-On 2: Run with curl
55 - 65	Overview 3: How RAG Works
65 - 85	Hands-On 3: Build a RAG Chatbot
85 - 90	Summary, OpenWebUI, and Next Steps

How This Class Works

Use the supporting website during the exercises. Keep the slides open to see what you are working on.

Slides and Supporting Website

- Slides: explanations and the current task
- Website: detailed steps, commands, and extra challenges
- Refer to the website while working

Required and Optional

- Each hands-on exercise has a required task and an optional challenge
- After completing the required task, continue to the optional challenge
- Work at your own pace

If You Get Stuck

- Search for the error message in Troubleshooting
- Ask a nearby classmate for help
- If the issue remains, ask the instructor or a TA

Overview 1 (0–10 min)

What Is SAKURA AI Engine?

What Is Generative AI?

Generative AI learns from large amounts of text and creates answers by predicting what comes next.



Different from Search

- Search finds information that already exists
- Generative AI creates an answer

Responses May Vary

- The wording can change even when you ask the same question

It Cannot Answer What It Does Not Know

- It cannot answer from internal documents or university-specific rules it has never received

This limitation leads directly to RAG, which we will use later in the class.

What Is SAKURA AI Engine?

SAKURA AI Engine is a secure and flexible API platform for building generative AI applications.

Multiple Foundation Models

- Choose the model that best fits your use case

Document Management and RAG

- Search registered documents and generate answers from their contents

Audio and Image Capabilities

- Audio Transcription, TTS (Text-to-Speech), and image input

Processing in Japan

- Data processing is completed in data centers in Japan

Four Major Functions

Chat Completions / Document Management and RAG / Audio Transcription / TTS

This class focuses on Chat Completions and Document Management and RAG.

Understanding the Concept of a Model

A model is the brain of an AI system. Models differ in what they have learned, their size, and what they are good at.

Smaller Model

- gemma-4-31B-it (31B)
- Lightweight and fast (Public Preview)

Standard Model

- gpt-oss-120b (120B)
- Used in this class

Larger Model

- Kimi-K2.7-Code
- High performance (Public Preview)

- The numbers indicate the parameter count, or model size
31B means approximately 31 billion; 120B means approximately 120 billion
- Available models may change. Check Available Models for the latest information.

In Hands-On 1, you will ask different models the same question and compare their answers.

Using AI in a Browser / Using AI via API

There are two ways to use the same AI. You will try both in this class.

Use It in a Browser (Playground)

- Enter a question on the screen and view the answer immediately
- No setup is required, so you can try it right away
- Suitable for comparing models and prompts

Used in Hands-On 1

Use It via API (Program)

- Call AI from a program
- Integrate AI into your own application or service
- Automate tasks and process large volumes of requests

Used in Hands-On 2

MISSION 01

Try SAKURA AI Engine

Required

1. Log in to the SAKURA Cloud Control Panel
2. Open SAKURA AI Engine
3. Review the list of Available Models
4. Select a model in Playground and ask a question

Optional | Extra Challenge

Ask the same question using another model and compare the answers.

Detailed steps and commands are on the supporting website. Keep this slide open while you work.

Before You Start: Check the Prerequisites

Hands-On 1 | Try SAKURA AI Engine

Before using SAKURA AI Engine, make sure the following items are ready.

Login Information (3 Items)

- User Code
- Member ID
- Password

SAKURA Cloud Project

- Create or select the project you will use for the exercises

Before You Continue

A User Code is also required for login. Make sure all three login items are available.

Control Panel: <https://secure.sakura.ad.jp/cloud/>

STEP 1 | Log In to the Control Panel

Hands-On 1 | Try SAKURA AI Engine

`https://secure.sakura.ad.jp/cloud/`

1 Open the URL above in your browser

2 Enter your User Code, Member ID, and Password
All three fields are required.

3 Confirm that the home screen appears

The screenshot shows the SAKURA internet login page. At the top, there are logos for SAKURA internet, SAKURA CLOUD, and a note about a dedicated server for PHY. Below the logos, the text "Log in" is displayed. There are two main login panels. The left panel is titled "Logged in as a Sakura Internet member" and contains fields for "Member ID" (with the example "nnn12345"), "password", and a checkbox for "Save Member ID". It has a blue "Log in" button and links for "Forgot your Member ID? Click here." and "Forgot your password? Click here.". The right panel is titled "Logged in as a Sakura Cloud user" and contains fields for "User code" (with the example "sakuraha-ai"), "Member ID" (with the example "nnn12345" and an "@" symbol), and "password". It has a green "Log in" button and a checkbox for "Save User Code and Member ID". Below the login panels, there is a link for "New Member Registration" for those who do not have a Sakura Internet member ID. The footer of the page says "© SAKURA internet inc."

If You Cannot Log In

A missing User Code is a common cause. Check that all three fields are filled in.

Paste the values carefully and make sure there are no extra spaces before or after them.

STEP 2 | Open SAKURA AI Engine

Hands-On 1 | Try SAKURA AI Engine

- 1

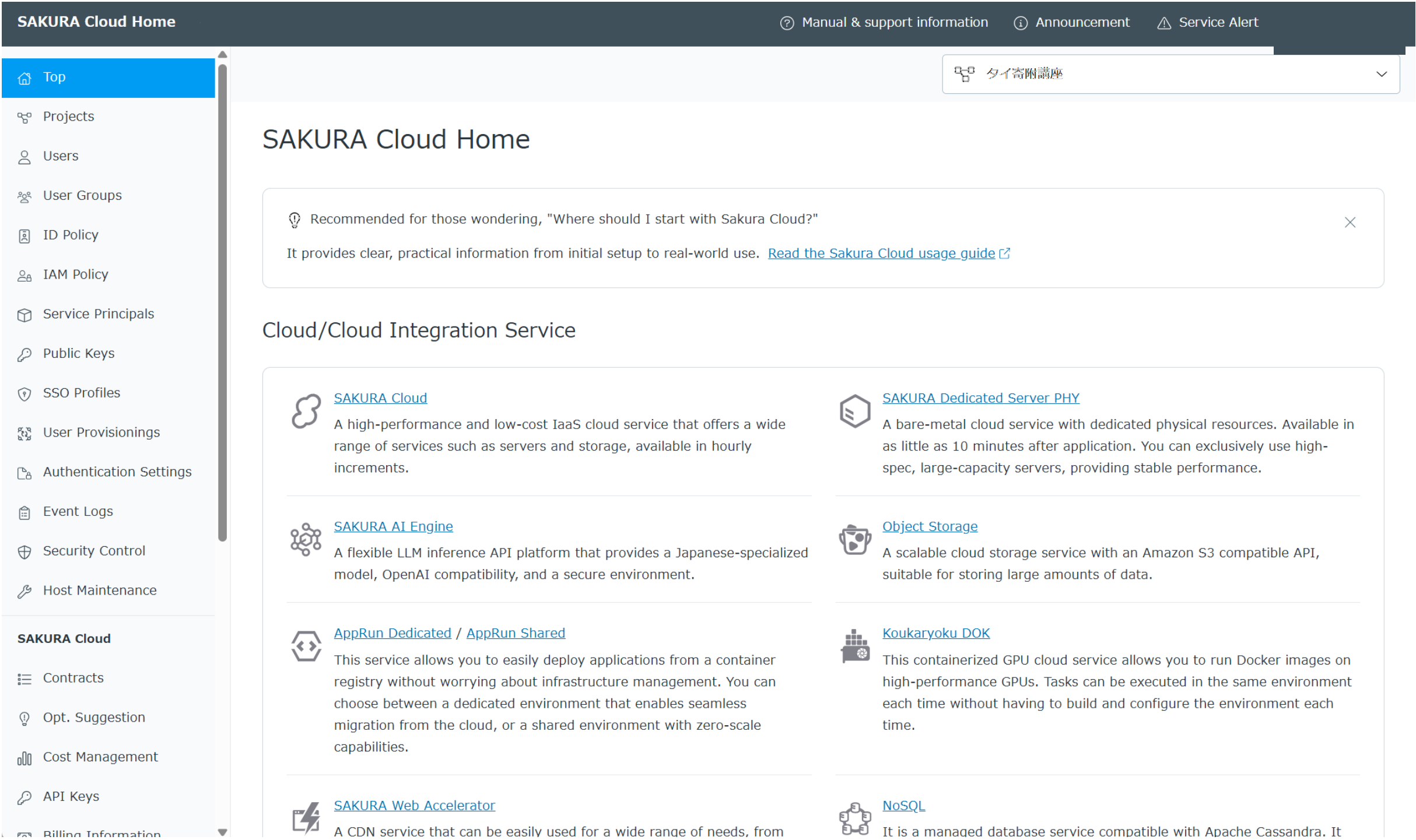
Open the menu on the home screen
- 2

Select SAKURA AI Engine from the service list
- 3

The step is complete when the SAKURA AI Engine screen appears

If the Service Is Not Listed

A project may not be selected. Select the correct project at the top of the screen.



SAKURA AI Engine Screen Overview

Hands-On 1 | Try SAKURA AI Engine

Available Models

View the models currently available

Account Tokens

Manage tokens used to access the APIs

Documents (RAG)

Manage documents used with RAG

Playground

Try models directly in your browser

Usage Amount

Review usage and charges

The four items highlighted in pink are used in this class.

To View Audio Models

Open Available Models and switch to the audio tab.

Available models

Account token

Document (RAG)

Usage amount

Playground

Available models (12)

Chat generation

Vector embedding

Audio transcription

Text-to-speech

For instructions on how to use the service, please refer to [the user manual](#).

Models with a status of "Application Required" are not available to individual customers. If you wish to use this service, please switch to a corporate or organizational account (not an individual account) and then apply.

Status: All

Tags: All

Filter by model name

Model Name	tag	status	Input fee	Output fee
Qwen3-Coder-30B-A3B-Instruct	coding	End of offe		
Qwen3-Coder-480B-A35B-Instruct-FP8	coding	End of offe		
gpt-oss-120b		available..	¥0.15 / 10,000 tokens (tax included)	¥0.75 / 10,000 tokens (tax included)
llm-jp-3.1-8x13b-instruct4	domestic	available..	¥0.15 / 10,000 tokens (tax included)	¥0.75 / 10,000 tokens (tax included)
preview/Kimi-K2.6	Public Preview multimodal +3	available..	¥0.60 / 10,000 tokens (tax included)	¥3.00 / 10,000 tokens (tax included)
preview/Kimi-K2.7-Code	Public Preview multimodal +3	available..	¥0.52 / 10,000 tokens (tax included)	¥5.04 / 10,000 tokens (tax included)

STEP 3 | Select a Model in Playground

Hands-On 1 | Try SAKURA AI Engine

- 1

Select Playground from the left menu
- 2

Open the model selector at the bottom of the screen
- 3

Select a model
Example: gpt-oss-120b

What You Can Do in Playground

- View the response to your prompt immediately
- Compare output differences and tendencies across models

 さくらのAI Engine playground

Enter your prompt here.

 gpt-oss-120b   Document 

STEP 4 | Enter and Submit a Prompt

Hands-On 1 | Try SAKURA AI Engine

1 Enter text in the prompt field

2 Click Submit or press Enter

3 Confirm that a response appears

Start with This Question

Please recommend three famous
tourist attractions in Thailand.

さくらAI Engine playground

Please recommend three famous tourist attractions in Thailand.

Sure! Here are three iconic Thai destinations that are must-visits for first-time travelers (and even seasoned tourists):

#	Attraction	Where	Why It's Famous	Quick Tips
1	Grand Palace & Wat Phra Kaew	Bangkok	The dazzling ceremonial complex that once housed the Thai king and the revered Emerald Buddha. Its intricate murals, golden spires, and the legendary "Chakri Maha Prasat" throne hall make it a showcase of Thai art and architecture.	• Dress modestly (no shorts, sleeveless tops, or flip-flops). • Arrive early (opens 8:30 am) to beat the crowds and heat. • Combine the visit with a short walk to nearby Wat Pho (home of the giant Reclining Buddha).
2	Doi Suthep (Wat Phra Prang)	Chiang Mai (mountain)	Perched on a 1,073-meter hill, this golden-crowned temple offers panoramic views of Chiang Mai and a sacred atmosphere. The 306-step staircase (or funicular) leads to the summit.	• Go at sunrise or late afternoon for the best light and fewer tourists. • Wear comfortable shoes.

Enter your prompt here.

gpt-oss-120b

Document

Once a response appears, you have completed the required part of Hands-On 1.

Available Models and Tags

Hands-On 1 | Try SAKURA AI Engine

Coding

- Models optimized for code generation

Multimodal

- Models that can handle multiple input formats, including images

Public Preview

- Provided for testing and feedback collection

Available models

Account token

Document (RAG)

Usage amount

Playground

Available models (12)

[Model licensing information](#)

[Chat generation](#) | [Vector embedding](#) | [Audio transcription](#) | [Text-to-speech](#)

For instructions on how to use the service, please refer to [the user manual](#).

利用可能

Tags: All

Filter by model name

Conditions met

available

End of offer

	tag	status	Input fee	Output fee
gpt-3.5-turbo		available..	¥0.15 / 10,000 tokens (tax included)	¥0.75 / 10,000 tokens (tax included)
llm-jp-3.1-8x13b-instruct4	domestic	available..	¥0.15 / 10,000 tokens (tax included)	¥0.75 / 10,000 tokens (tax included)
preview/Kimi-K2.6	Public Preview multimodal +3	available..	¥0.60 / 10,000 tokens (tax included)	¥3.00 / 10,000 tokens (tax included)
preview/Kimi-K2.7-Code	Public Preview multimodal +3	available..	¥0.52 / 10,000 tokens (tax included)	¥5.04 / 10,000 tokens (tax included)
preview/Phi-4-mini-instruct-cpu	Public Preview CPU	available..	¥0.01 / 10,000 tokens (tax included)	¥0.03 / 10,000 tokens (tax included)
preview/Qwen3-0.6B-cpu	Public Preview CPU	available..	¥0.01 / 10,000 tokens (tax included)	¥0.03 / 10,000 tokens (tax included)

Tags help you identify the use cases for which each model is suited.

Optional | Compare Different Models

Hands-On 1 | Try SAKURA AI Engine

Ask the same question using another model. Compare the length, style, and content of the answers.

What to Compare

- Which model is best suited to each use case
- How changes to the prompt affect the response

gpt-oss-120b

 playground

Please recommend three famous tourist attractions in Thailand.

Sure! Here are three iconic Thai destinations that are must-visits for first-time travelers (and even seasoned tourists):

#	Attraction	Where	Why It's Famous	Quick Tips
1	Grand Palace & Wat Phra Kaew	Bangkok	The dazzling ceremonial complex that once housed the Thai king and the revered Emerald Buddha. Its intricate murals, golden spires, and the legendary "Chakri Maha Prasat" throne hall make it a showcase of Thai art and architecture.	• Dress modestly (no shorts, sleeveless tops, or flip-flops). • Arrive early (opens 8:30am) to beat the crowds and heat. • Combine the visit with a short walk to nearby Wat Pho (home of the giant Reclining Buddha).
2	Doi Suthep (Wat Phra Prang)	Chiang Mai (mountain)	Perched on a 1,073-meter hill, this golden-crowned temple offers panoramic views of Chiang Mai and a sacred atmosphere. The 306-step staircase (or funicular) leads to the temple.	• Go at sunrise or late afternoon for the best light and fewer tourists. • Wear comfortable shoes for the climb.

Enter your prompt here.

 gpt-oss-120b  Document 

llm-jp-3.1-8x13b-instruct4

 playground

Please recommend three famous tourist attractions in Thailand.

1. Bangkok: Bangkok is Thailand's capital and a bustling metropolis with rich culture and history. Some of the top attractions include the Grand Palace, Wat Pho (the Temple of the Reclining Buddha), and the Emerald Buddha Temple (Wat Phra Kaew). Other must-visit places are the vibrant street markets of Asiatique, the striking Saatchi Gallery, and the beautiful Lumpini Park.

2. Chiang Rai: Chiang Rai is known for its stunning natural beauty, and the famous White Temple (Wat Rong Khun) is a popular attraction. The temple's unique design, covered completely in white plaster and mirror fragments, reflects the belief in the purity of Buddhism. Other attractions include the Black House, an art museum and architectural showcase, and the Golden Triangle, where Thailand, Laos, and Myanmar meet.

3. Phuket: As Thailand's largest island, Phuket offers a blend of stunning beaches, vibrant nightlife, and rich culture. Some of the top attractions include the beautiful beaches of Patong, Kata Noi, and Karon, the Phuket Old Town, and the Thai-Buddhist temples of Wat Chalong and Wat Sri Sunthon. Additionally, the Phuket Fantasea show offers a great family-friendly cultural experience with its stunning performances and delicious dinner.

Enter your prompt here.

 llm-jp-3.1-8x13b-instruct4  Document 

Progress Check

Checkpoint

Before moving on, confirm that you have completed each item.

- ✓ **Logged in to SAKURA Cloud**
- ✓ **Opened the SAKURA AI Engine screen**
- ✓ **Selected a model and asked a question in Playground**
- ✓ **Confirmed that the response appeared**

How to Proceed

If you are finished, continue to the optional challenge. If you are stuck, ask the instructor or a TA.

Playground Uses the API Behind the Scenes

Everything you did in Playground was actually performed through the API.



Everything you did in Playground also uses the API behind the scenes.

Use the Browser Interface (Playground)

- A person operates it one request at a time
- Quick to try

Use the Command Line (curl)

- Call the same API yourself
- Integrate the call into a program

Next, you will operate this behind-the-scenes process yourself.

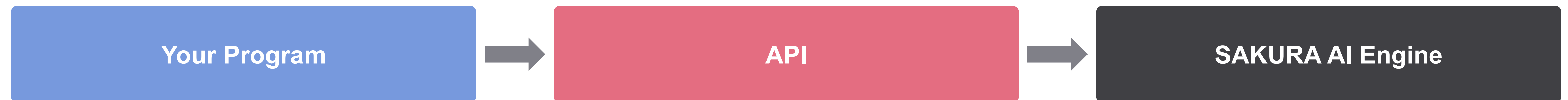
Overview 2 (25–35 min)

What Is an API?

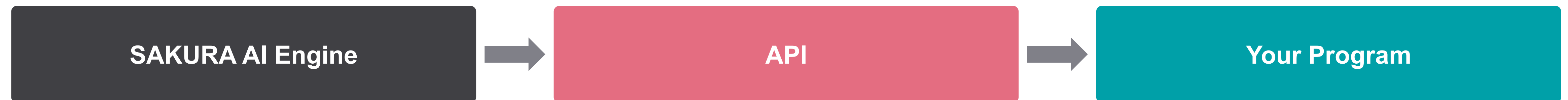
What Is an API?

An API is an interface that allows a program to call a service.

1. Send a Question



2. Receive an Answer



Use a Defined Request and Response Format

- Send a request in a defined format and receive a response in a defined format

APIs Are Used by Programs

- Chat AI and web apps are also connected through APIs behind the scenes

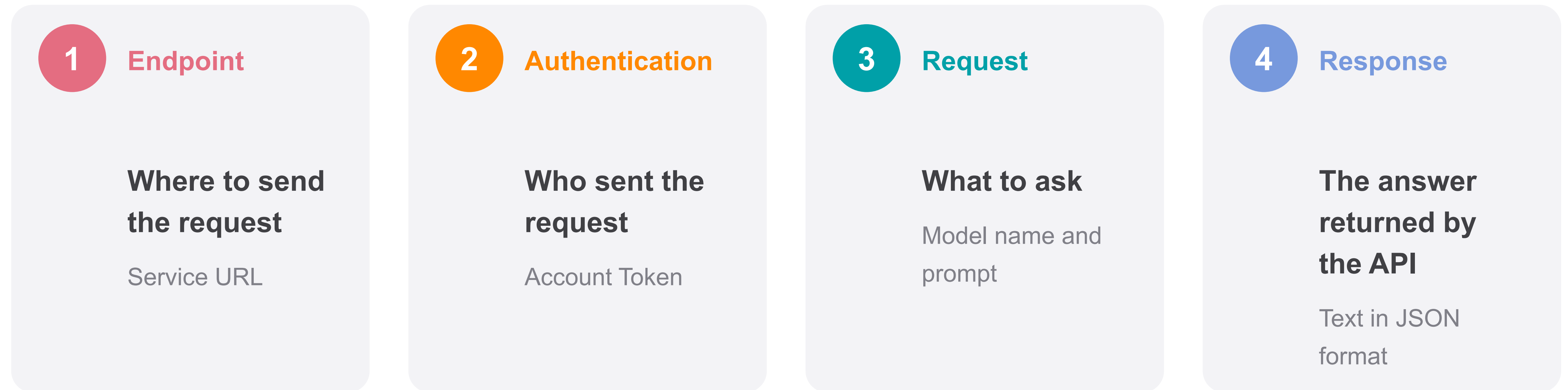
APIs Enable Integration

- You can integrate AI into your own application

In Hands-On 2, you will call this API from the command line using curl.

What You Need to Define When Using an API

You can call AI once these four items are defined.



In Hands-On 2, you will identify where these four items appear in the curl command.

MISSION 02

Run AI from the Command Line

Required

1. Issue an Account Token
2. Store the token in an environment variable
3. Send a question to SAKURA AI Engine using curl
4. Extract the answer from the returned JSON
5. Edit the prompt and run the command again

Optional | Extra Challenge

- Change output conditions and parameters
- Call the same API from Python

Copy the commands from the supporting website. Typing them manually can introduce errors.

STEP 1 | Issue an Account Token

Hands-On 2 | Run AI from the Command Line

- 1

Select Account Tokens from the left menu
- 2

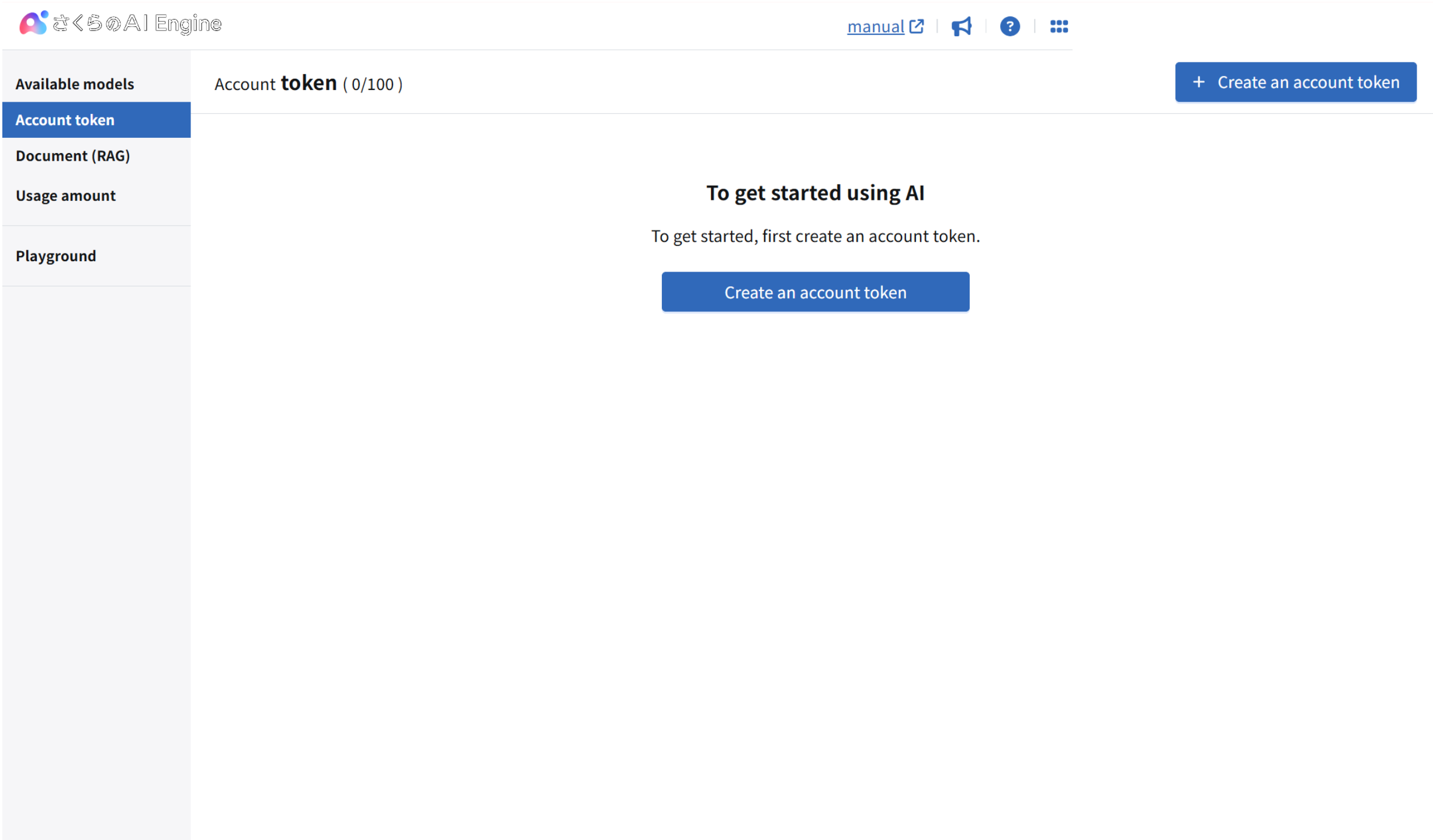
Click Create Account Token
- 3

Enter an Account Token Name
Example: rag-sample-token
- 4

Copy and securely save the displayed token

Important

The token is displayed only when it is issued. Copy and save it before closing the completion screen.



Account Token Format and Handling

Hands-On 2 | Run AI from the Command Line

Copy and securely save the token as soon as it is displayed.

Token Format (Example)

<UUID>:<secret>

UUID

- Access token ID

Secret

- A secret string displayed only once

Security Precautions

- Do not share or publish the token
- If you lose it, issue a new token
- Delete tokens that are no longer needed

We Will Now Use the Terminal

Hands-On 2 | Run AI from the Command Line

From this point, use WSL (Ubuntu) on Windows. This course assumes WSL (Ubuntu).

Start WSL

- Open Ubuntu from the Start menu
- Or select Ubuntu in Windows Terminal
- Or open PowerShell and type wsl

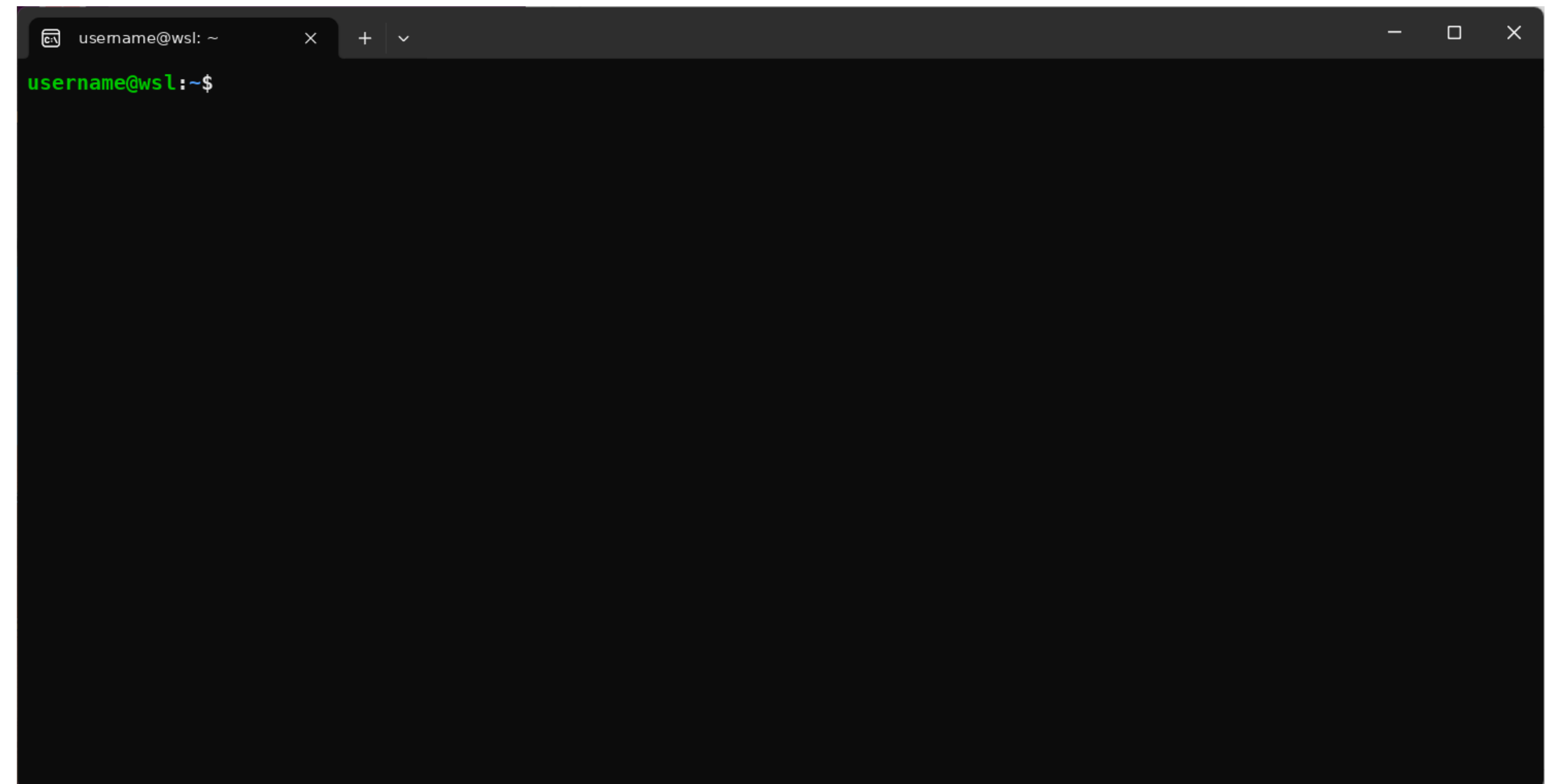
Why Use WSL?

The commands in this material assume a Linux environment.

WSL lets you run Linux commands while using Windows.

Reference Environment Used to Test This Material

Ubuntu 22.04.5 LTS / Python 3.10.12 / Node.js v24.11.0



STEP 2 | Store the Token in an Environment Variable

Hands-On 2 | Run AI from the Command Line

Store the token in a variable first so that you do not need to paste it into every command.

1. Store the Token in an Environment Variable

```
export AI_ENGINE_TOKEN="<issued account token>"
```

2. Confirm That It Was Set Correctly

```
echo $AI_ENGINE_TOKEN  
# The setup is correct if the issued token is displayed.
```

Remember

This setting remains active until you close the terminal. After reopening the terminal, run export again.

STEP 3 | Ask AI a Question with curl

Hands-On 2 | Run AI from the Command Line

```
curl --location https://api.ai.sakura.ad.jp/v1/chat/completions ¥  
--header 'Accept: application/json' ¥  
--header "Authorization: Bearer ${AI_ENGINE_TOKEN}" ¥  
--header 'Content-Type: application/json' ¥  
--data '{  
  "model": "gpt-oss-120b",  
  "messages": [  
    { "role": "user", "content": "Hello!" }  
  ],  
  "temperature": 0.7,  
  "max_tokens": 200,  
  "stream": false  
}'
```

1

Endpoint

Where to send the request

2

Authentication

Pass the token stored in the environment variable

3

Request

Model name and question

4

Response

The response appears in the terminal after execution

curl is a command for sending and receiving data through a URL. Here, it sends a question to AI from the command line instead of a browser.

Request Parameters

Hands-On 2 | Run AI from the Command Line

Parameter	Meaning
model	Name of the AI model to use
messages	Conversation content, including roles and messages
role	Speaker role: user for user input; system for AI instructions
content	Message text to send. Edit this field.
temperature	Controls response diversity. Lower is more consistent; higher is more varied.
max_tokens	Maximum number of tokens in the generated response
stream	Response mode: streaming or non-streaming

How to Read a JSON Response

Hands-On 2 | Run AI from the Command Line

```
{
  "id": "chatcmpl-xxxxxxx",
  "object": "chat.completion",
  "model": "gpt-oss-120b",
  "choices": [
    {
      "index": 0,
      "message": {
        "role": "assistant",
        "content": "Hello! How can I help you today?"
      },
      "finish_reason": "stop"
    }
  ],
  "usage": { "prompt_tokens": 12,
             "completion_tokens": 20 }
}
```

1

The answer is stored here

2

stop means generation completed normally

3

Number of tokens used, which affects usage charges

What Is JSON?

JSON is a format that programs use to exchange data.

You can read it as a collection of name-and-value pairs inside { }.

Extract Only the Answer

Hands-On 2 | Run AI from the Command Line

You can display only the answer text instead of the entire JSON response.

Extract with Python (No Additional Installation Required)

```
curl -s --location https://api.ai.sakura.ad.jp/v1/chat/completions ¥  
  --header "Authorization: Bearer ${AI_ENGINE_TOKEN}" ¥  
  --header 'Content-Type: application/json' ¥  
  --data '{ "model": "gpt-oss-120b",  
            "messages": [{ "role": "user", "content": "Hello!" }] }' ¥  
| python3 -c "import sys,json; print(json.load(sys.stdin)['choices'][0]['message']['content'])"
```

What Is a Pipe (|)?

A pipe passes the output of curl directly to the next command.

Here, the returned JSON is passed to Python, which extracts only the answer text.

STEP 4 | Edit the Prompt and Run Again

Hands-On 2 | Run AI from the Command Line

Edit messages and run the command again

```
--data '{
  "model": "gpt-oss-120b",
  "messages": [
    { "role": "system", "content": "You are a helpful travel guide." },
    { "role": "user", "content": "Recommend three famous tourist attractions in Thailand." }
  ],
  "max_tokens": 300
}'
```

"role": "system"

- Instructions that define how the AI should behave

"role": "user"

- The user message containing the question

Once the edited request runs successfully, you have completed the required part of Hands-On 2.

Optional | Change the Response Style and Parameters

Hands-On 2 | Run AI from the Command Line

Add Conditions to the Question

```
"content": "Recommend three famous tourist attractions in Thailand.  
Use bullet points. Less than 100 words."
```

Use bullet points.

- Request a bulleted answer

Less than 100 words.

- Limit the answer to 100 words

Write for high school students.

- Request clear, accessible language

"temperature": 0.2 / 1.2

- Lower values produce more consistent answers
- Higher values produce more varied answers

"max_tokens": 50 / 500

- Sets the maximum response length
- A small value may cut the answer off

Optional | Call the API from Python

Hands-On 2 | Run AI from the Command Line

Call the same API from a program.

chat.py

```
import os, requests

url = "https://api.ai.sakura.ad.jp/v1/chat/completions"
headers = {
    "Authorization": f"Bearer {os.environ['AI_ENGINE_TOKEN']}",
    "Content-Type": "application/json",
}
payload = {
    "model": "gpt-oss-120b",
    "messages": [{"role": "user", "content": "Hello!"}],
    "max_tokens": 200,
}
res = requests.post(url, headers=headers, json=payload)
print(res.json()["choices"][0]["message"]["content"])
```

1

Save the file as chat.py

2

Install requests only if it is not already installed

3

Run the program

The Same Process as curl

Both approaches send the same three elements: a URL, headers, and data.

Common Errors and Solutions

Hands-On 2 | Run AI from the Command Line

An error does not mean failure. The message usually indicates the cause.

401 Unauthorized

The token was not passed correctly

- Check the value with echo
\$AI_ENGINE_TOKEN
- Check for extra spaces before or after the token
- Check that there is a single space after Bearer

The Command Stops Before Completion

The quotation marks are full-width characters

- Use " instead of ”
- Turn off Japanese input before pasting
- Do not add a space after the backslash at the end of a line

Cannot Connect / No Response

A network or proxy issue

- Check that Wi-Fi is connected
- Check whether the campus network has restrictions

Search for the exact error message in Troubleshooting on the supporting website.

Progress Check

Checkpoint

Before moving on, confirm that you have completed each item.

- ✓ **Issued and saved an Account Token**
- ✓ **Stored the token in an environment variable**
- ✓ **Received an AI response using curl**
- ✓ **Edited the prompt and ran the command again**

How to Proceed

If you are finished, continue to the optional challenge. If you are stuck, ask the instructor or a TA.

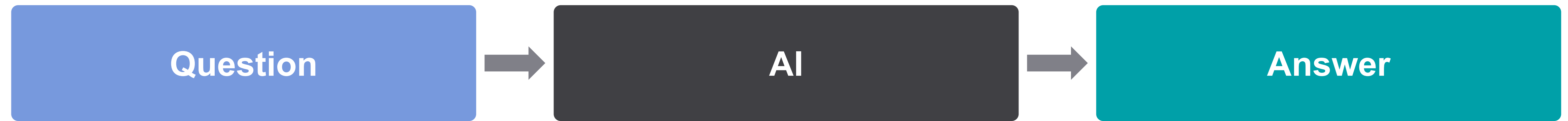
Overview 3 (55–65 min)

How RAG Works

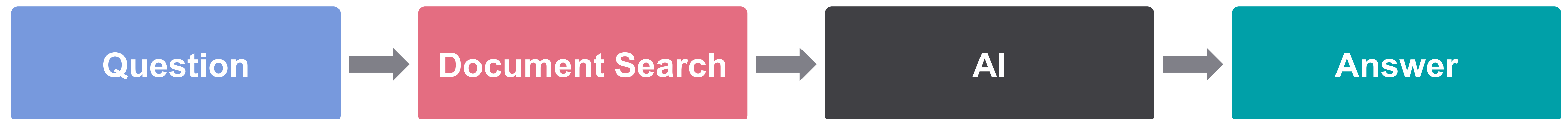
What Is RAG?

RAG adds one key step: searching registered documents before generating a response.

Without RAG



RAG



RAG stands for Retrieval-Augmented Generation. It is an AI approach that answers based on registered documents.

What RAG Can and Cannot Do

RAG is not universal. Its evidence is limited to the documents that have been registered.

What It Can Do

- Answer based on registered documents
- Answer questions about information a general AI does not know
- Produce different answers when the documents are replaced
- Show which documents were referenced

What It Cannot Do

- Use unregistered information as evidence
- Correct errors in the source document automatically
- Fall back on general knowledge outside the documents
- Continue using a document after it is deleted

In Hands-On 3, you will intentionally ask a question that is not answered in the document.

MISSION 03

Build a RAG Chatbot

Required

1. Download and register the provided PDF
2. Ask a question without RAG and record the answer
3. Ask the same question with RAG and record the answer
4. Ask a question that is not covered by the document
5. Describe the difference in one sentence on the worksheet

Optional | Extra Challenge

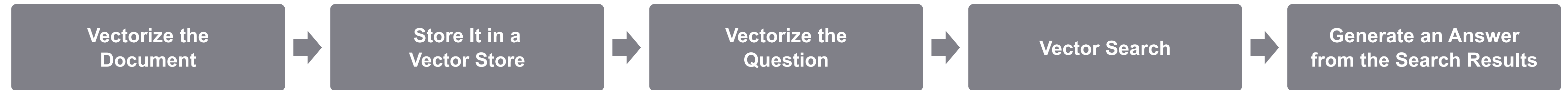
- Try your own PDF
- Register a document and run RAG through the API
- Change `top_k` and `threshold` and compare the answers

Detailed steps and commands are on the supporting website. Keep this slide open while you work.

RAG in SAKURA AI Engine

Hands-On 3 | Build a RAG Chatbot

Normally, building RAG requires you to implement the following five processes yourself.



With SAKURA AI Engine

You do not need to implement these processes yourself.

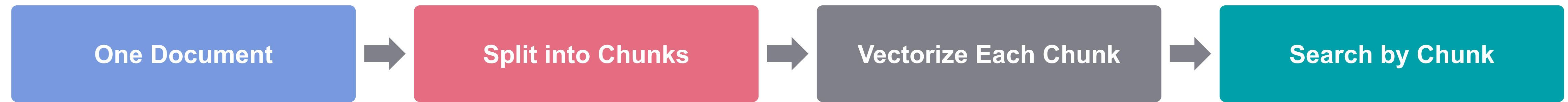
From vectorizing and storing registered documents to searching and generating answers, the entire workflow can be performed through API calls.

First, try it in the Control Panel without code, and then run it through the API.

Understanding Chunks

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Registered documents are divided into sections of a specified token length. Each section is called a chunk.



Searches Are Performed by Chunk

- Multiple chunks may be returned

Only Relevant Sections Are Referenced

- The entire document is not sent to the model

Supported File Formats

txt / pdf / html / docx / xlsx / md and other common document formats

Important: Billing Notice Before Registration

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Registering Documents Incurs Charges

- Monthly charges based on the number of chunks continue until the document is deleted
- These charges also apply when using the Foundation Model Free Plan
- The number of chunks depends on the chunk size and source text length

In This Class

- You will use a training environment
- You will not be charged directly

In Actual Use

- Monthly charges depend on the amount of registered document data
- Remember that charges are incurred every month

Make a habit of deleting documents after use to avoid unnecessary costs.

STEP 1 | Prepare the Document

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Download the document from the supporting website. Everyone will use the same PDF.

Required: Provided PDF

- Download the PDF from the supporting website

Where to Find the Document

- Open the Hands-On 3 page on the supporting website
- Use the download link on that page

Optional: Your Own PDF

- You may also try a syllabus or student handbook

Why Use the Provided Document?

Using the same document and questions makes it easier to compare results with and without RAG.

Next

Place the downloaded PDF in the WSL folder by following the steps on the next slide.

STEP 1 Continued | Place the Document in a WSL Folder

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Copy the file using File Explorer, not a command.

1 Open File Explorer
From the left pane, select Linux > Ubuntu > home

2 Create a folder named 01_AI_RAG

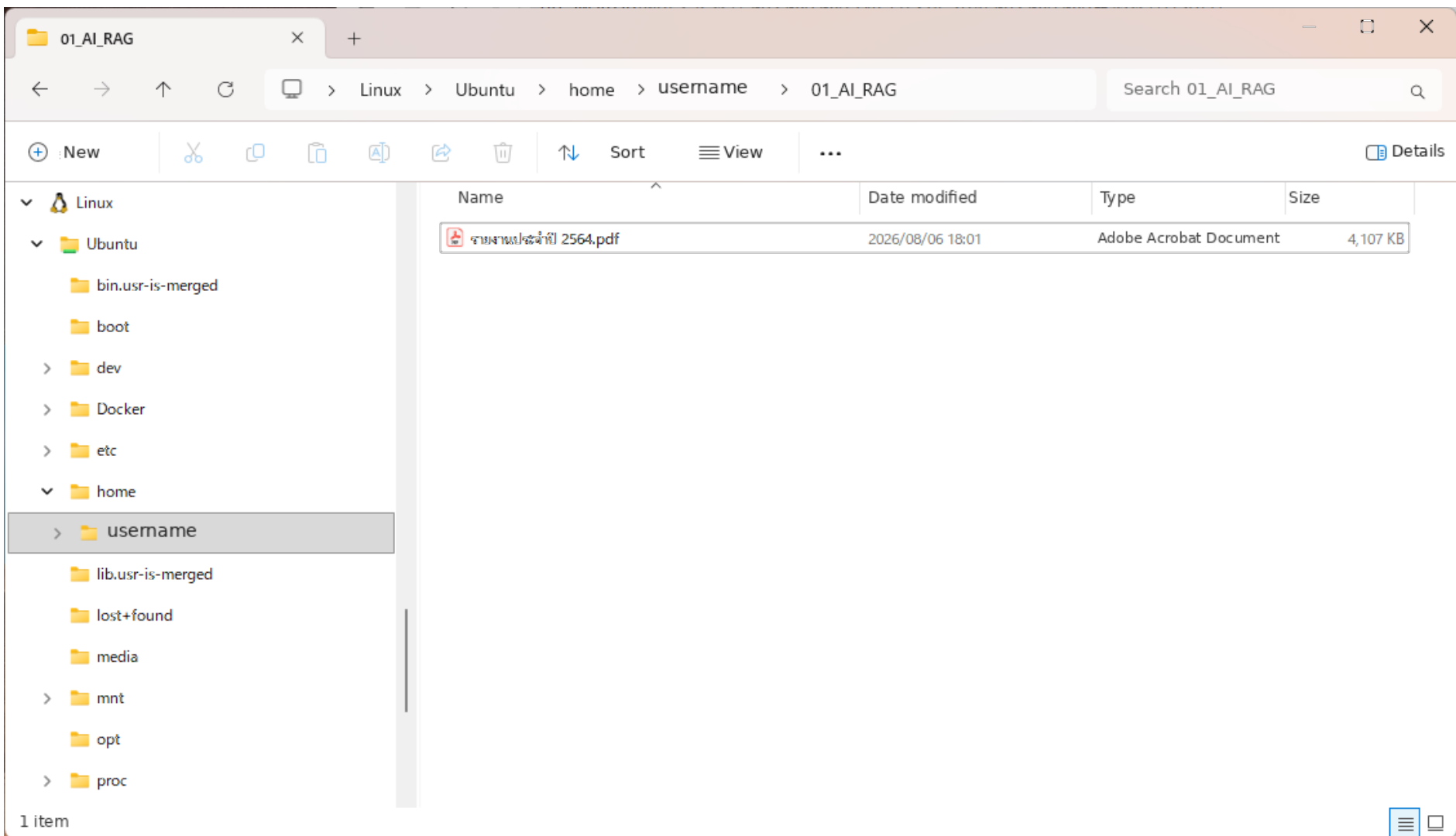
3 Copy the downloaded PDF
You can drag and drop it

4 Return to WSL and confirm the file

If Linux Is Not Listed

Enter the following path in the address bar:

\\wsl\$\\Ubuntu\\home



Confirm in WSL (Ubuntu)

```
cd ~/01_AI_RAG
```

```
ls -l
```

The setup is correct if the PDF appears.

STEP 2 | Register a Document

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- 1

Select Documents (RAG) from the left menu
- 2

Click Add Document
- 3

Enter the required information and upload the file
- 4

Wait until the status changes from pending to available

Field	Content
Name	Title of the document
File	Document file to upload
Tags	Optional tags used to filter searches
Embedding Model	Model used for vectorization
Chunk Size	Maximum number of tokens in one chunk

Available models

Account token

Document (RAG)

Usage amount

Playground

manual

Note

Please note that once you register documents for use with RAG, a monthly fee will be charged based on the number of chunks in those documents until you delete them. This fee applies even if you are using the free plan of the base model.
Price: ¥3/100 chunk (tax included)

Please enter the document information.

name

Annuual report 2564

Document file

ファイルを選択

รายงานประจำปี 2564.pdf

tag

Add tags

Model

multilingual-e5-large

Chunk size

512

The chunk size range you can specify for the selected model is 256 to 512.

cancel

Add

Documents That Cannot Be Used or May Reduce Accuracy

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Not every document can be imported accurately.

Cannot Be Processed Reliably

- Image-only PDFs, such as scanned documents
- Encrypted or password-protected files
- Corrupted or unsupported files

May Reduce Accuracy

- Multi-column layouts
- Special fonts or vertical writing
- Documents with many equations, charts, or complex layouts

Remember

This service does not guarantee that every file will be imported completely.

If the RAG answer looks incorrect, inspect the source document and its format.

STEP 3 | Run a RAG Query

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- 1

Open Documents (RAG) from the left menu
- 2

Click Operation in the upper-right corner
- 3

Select Query Search
- 4

Enter the question below and run the search

You can also specify tags, the model, and the number of chunks

How to Read the Results

- A natural-language answer is displayed
- The referenced document is shown

Question for Everyone

Explain KOSEN-KMITL and Monozukuri.

Available models

Account token

Document (RAG)

Usage amount

Playground

search

Query results

answer

KOSEN-KMITL

The material mentions a cooperation between **KOSEN** (the Japanese system of “National Colleges of Technology”) and **KMITL** (King Mongkut's Institute of Technology Ladkrabang, Thailand). In the strategic plan that integrates four projects, KOSEN experts are invited to help create new **innovation ecosystems** and to support **startup** and **deep-tech** development in Thailand. The partnership is described as a way to “เชื่อมต่อการนวัตกรรม” (connect innovation) and to “ส่งเสริม Startup” together with leading Thai companies, thereby strengthening KMITL's role as a technology-and-innovation institute that can compete nationally and internationally.

Monozukuri

“Monozukuri” appears in the description of **Project 2: Monozukuri Degree** . The Monozukuri-based degree is aimed at **developing basic (พื้นฐาน) skills** and providing **expertise in technology and entrepreneurship** in fields such as **electrical engineering, automotive engineering, and Mechatronics** . It follows the Japanese “Monozukuri” philosophy, which emphasizes **craftsmanship, manufacturing excellence, and the integration of technical know-how with entrepreneurial ability** . The program is intended to produce graduates who can **create and run new-generation enterprises (Young Entrepreneurs)** and compete on a global scale.

In short

KOSEN-KMITL is a collaborative link between Japanese technical colleges (KOSEN) and Thailand's KMITL to boost innovation, deep-tech startups, and international competitiveness. **Monozukuri** is a Japanese-origin training model adopted by KMITL to build human-capital skills in engineering and entrepreneurship, forming the backbone of the “Monozukuri Degree” program.

(Any details not explicitly stated in the provided excerpts are **unknown**)

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STEP 4 | Use RAG in Playground

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- 1

Open Playground
- 2

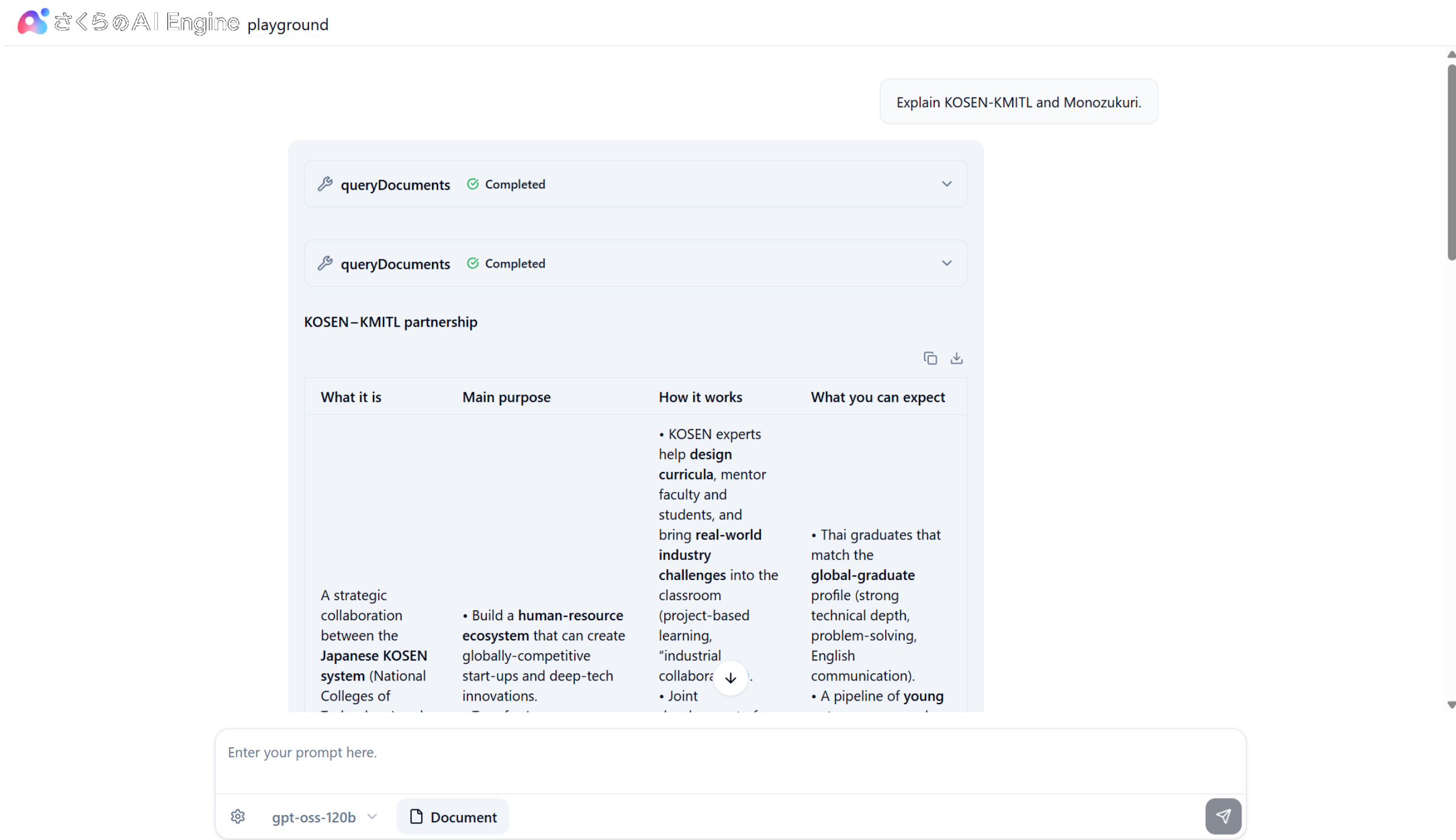
Click the Document button at the bottom of the screen

This enables document search
- 3

Set the parameters using the gear icon in the lower-left corner
- 4

Enter a question and click Submit

Parameter	Meaning
tags	Reference only documents with the selected tags
top_k	Maximum number of top-ranked chunks to retrieve
threshold	Vector-distance threshold (0–2)



STEP 5 | Compare Results Without and With RAG

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The key is to try the same question both without and with RAG.

1

Ask Question 1 without RAG

Turn the Document button off and record the answer

2

Ask Question 1 with RAG

Turn the Document button on and record the answer

3

Ask Question 2 with RAG

This question is not covered by the document

4

Describe the difference in one sentence

Write it on the worksheet

What to Observe

- Does the answer to Question 1 reflect the document?
- Where does the answer to Question 2 come from?

Question 2 Is the Most Important

RAG can only use registered documents as evidence.

Confirm this for yourself by observing the result.

Questions for Everyone

- (1) Explain KOSEN-KMITL and Monozukuri.
- (2) Who will win the next FIFA World Cup?

Compare Results Without and With RAG

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These are two answers to the same question: “Explain KOSEN-KMITL and Monozukuri.”

WITHOUT RAG (Document button off)

さくらのAI Engine playground

Explain KOSEN-KMITL and Monozukuri.

1. What is KOSEN-KMITL?

1.1 The two partners

Institution	What it is	Core strength
KOSEN (National Institute of Technology, Japan)	A network of 60 "colleges of technology" that offer 5-year "Kōsen" programs (high-school-level + junior-college) for young engineers.	Very practice-oriented curricula, strong ties with industry, high graduation-to-employment rates, a culture of monozukuri (craftsmanship) and kaizen (continuous improvement).
KMITL (King Mongkut's Institute of Technology Ladkrabang, Thailand)	One of Thailand's premier universities of engineering, science and technology.	Large research campus, advanced facilities (e.g., robotics labs, smart-factory testbeds), and a national mandate to raise Thailand's "Industry 4.0" capability.

1.2 Why the partnership was created

Enter your prompt here.

gpt-oss-120b

Document

WITH RAG (Document button on)

さくらのAI Engine playground

Explain KOSEN-KMITL and Monozukuri.

queryDocuments Completed

queryDocuments Completed

KOSEN – KMITL partnership

What it is	Main purpose	How it works	What you can expect
A strategic collaboration between the Japanese KOSEN system (National Colleges of Technology)	• Build a human-resource ecosystem that can create globally-competitive start-ups and deep-tech innovations.	• KOSEN experts help design curricula , mentor faculty and students, and bring real-world industry challenges into the classroom (project-based learning, "industrial collabora...") • Joint	• Thai graduates that match the global-graduate profile (strong technical depth, problem-solving, English communication). • A pipeline of young

Enter your prompt here.

gpt-oss-120b

Document

With RAG, the answer is based on the registered document, and the referenced sources are available for review.

How to Complete the Worksheet

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The Hands-On 3 page on the supporting website includes fields for your answers.

Answer WITHOUT RAG

Copy the answer generated without RAG

Answer WITH RAG

Copy the answer generated with RAG

What Was the Difference?

Describe the difference in one sentence

Your entries are saved automatically and remain available after you reload the page.

Optional | Register a Document via API

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The same operation performed in the Control Panel can also be executed through the API.

1. Upload the Document (Run from ~/01_AI_RAG)

```
curl --request POST ¥  
  --url https://api.ai.sakura.ad.jp/v1/documents/upload/ ¥  
  --header 'Accept: application/json' ¥  
  --header "Authorization: Bearer ${AI_ENGINE_TOKEN}" ¥  
  --header 'Content-Type: multipart/form-data' ¥  
  --form "file=@<document-file-name>.pdf"
```

How to Read the Response

If "status": "pending" is returned, the document has been added to the processing queue.

After a short wait, the status becomes available and the document can be searched.

Optional | Check Document Status via API

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2. Extract Only the ID, Status, and Name

```
curl -s --location 'https://api.ai.sakura.ad.jp/v1/documents/<document-id>/' \
  --header "Authorization: Bearer ${AI_ENGINE_TOKEN}" \
  | grep -oE '"id": "[^"]*"|"status": "[^"]*"|"name": "[^"]*"'
```

pending

- Processing is pending. The document cannot be searched yet.

available

- Vectorization is complete. The document can now be searched.

What Does `grep -oE` Do?

It extracts only the parts of the output that match the specified pattern.

This lets you check only the necessary fields without reading the entire JSON response.

Optional | Run RAG via API

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3. Generate an Answer Grounded in the Document

```
curl --request POST ¥
  --url https://api.ai.sakura.ad.jp/v1/documents/chat/ ¥
  --header 'Accept: application/json' ¥
  --header "Authorization: Bearer ${AI_ENGINE_TOKEN}" ¥
  --header 'Content-Type: application/json' ¥
  --data '{
    "model": "multilingual-e5-large",
    "chat_model": "gpt-oss-120b",
    "query": "Explain KOSEN-KMITL and Monozukuri.",
    "top_k": 3,
    "threshold": 0.6,
    "distance_type": "cosine"
  }'
```

The Response Includes

- The generated answer
- Information about the referenced documents (sources)

Search and Generation in One Call

- A process that normally requires multiple steps
- can be completed with a single API call

Optional | Document / RAG API Parameters

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Parameter	Meaning
<code>model</code>	Model used for vectorization and embedding
<code>chat_model</code>	LLM used to generate the answer
<code>query</code>	Question in natural language
<code>top_k</code>	Maximum number of top-ranked chunks to retrieve
<code>threshold</code>	Vector-distance threshold. More distant content is excluded (0–2).
<code>distance_type</code>	cosine: cosine distance; l2: Euclidean distance

If `distance_type` is omitted, cosine distance is used.

Optional | Choose the Appropriate API

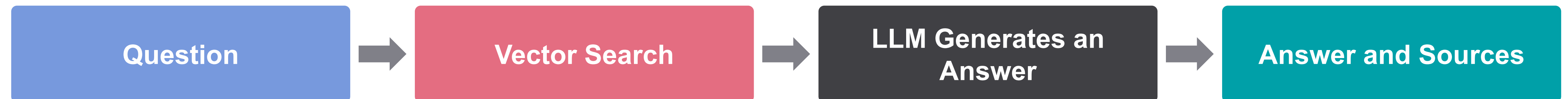
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/v1/chat/completions

- Calls an LLM without document search
- Does not search registered documents
- Use it for general conversations and text generation

/v1/documents/chat/

- Runs document search and answer generation in one call
- Generates an answer grounded in registered documents
- Use it when the answer should reference documents



The documents_query API performs document search only. Use it when you need to process the search results directly without generating a natural-language answer.

Optional | Change the Settings and Compare

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top_k

- Increase it to retrieve more chunks
- Decrease it to use only the highest-ranked chunks

For This Exercise, Change threshold

- The document is small, so changing top_k may not produce an obvious difference

threshold

- Increase it to use only more closely related content
- Decrease it to include less closely related content

What to Observe

- Does a lower threshold retrieve less relevant sections?
- How do the answer and wording change?

Another Optional Challenge

Delete the document and confirm that the answer returns to the result without RAG.

How Changing the Threshold Affects the Answer

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“threshold”: 0.6 | Use only closely related content

```
{
  "model": "multilingual-e5-large",
  "chat_model": "gpt-oss-120b",
  "query": "Explain KOSEN-KMITL and Monozukuri.",
  "top_k": 3,
  "threshold": 0.6,
  "distance_type": "cosine"
}
```

["answer": "**KOSEN-KMITL**\n\n- The material mentions **KOSEN** as a Japanese model of technology-focused higher-education institutes that “ช่วยสร้างนวัตกรรม” (helps create innovation) and works together with “บริษัทชั้นนำในประเทศ” (leading domestic companies) to promote startups and entrepreneurship. \n- In the “Innovative level” section the text says that KOSEN (from Jap

“threshold”: 0.3 | Include less closely related content

```
{
  "model": "multilingual-e5-large",
  "chat_model": "gpt-oss-120b",
  "query": "Explain KOSEN-KMITL and Monozukuri.",
  "top_k": 3,
  "threshold": 0.3,
  "distance_type": "cosine"
}
```

["answer": "**KOSEN (as it appears in the document)** \n- The text refers to **KOSEN** as a group of Japanese technical colleges (“KOSEN จากประเทศไทย”). \n- These institutions are mentioned as partners that will **help create innovation** and **support start-up development**. \n- The collaboration with KOSEN is part of a broader strategy to integrate expertise from l

Lowering the threshold retrieves additional parts of the document, which changes the answer.

Review

Check whether you can explain these three points in your own words.

Q1 How did the AI answer without RAG?

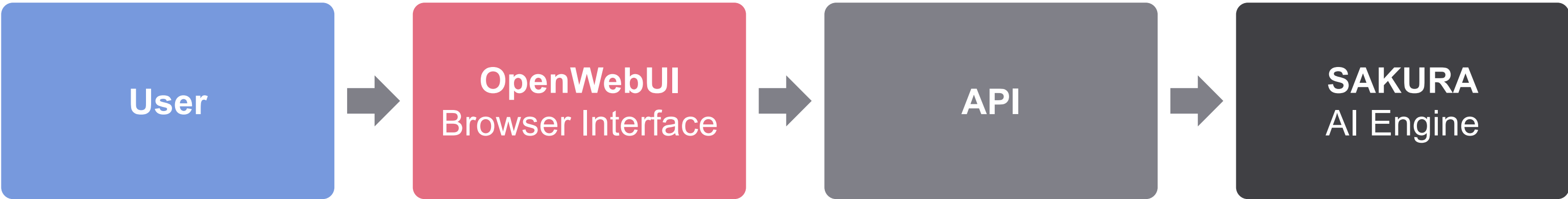
Q2 How did the answer change with RAG?

Q3 What happened when you asked about something not covered by the document?

RAG can only use registered documents as evidence. That is the key takeaway from this class.

Introducing OpenWebUI

OpenWebUI adds a ready-to-use interface on top of the same API workflow you used from the command line.



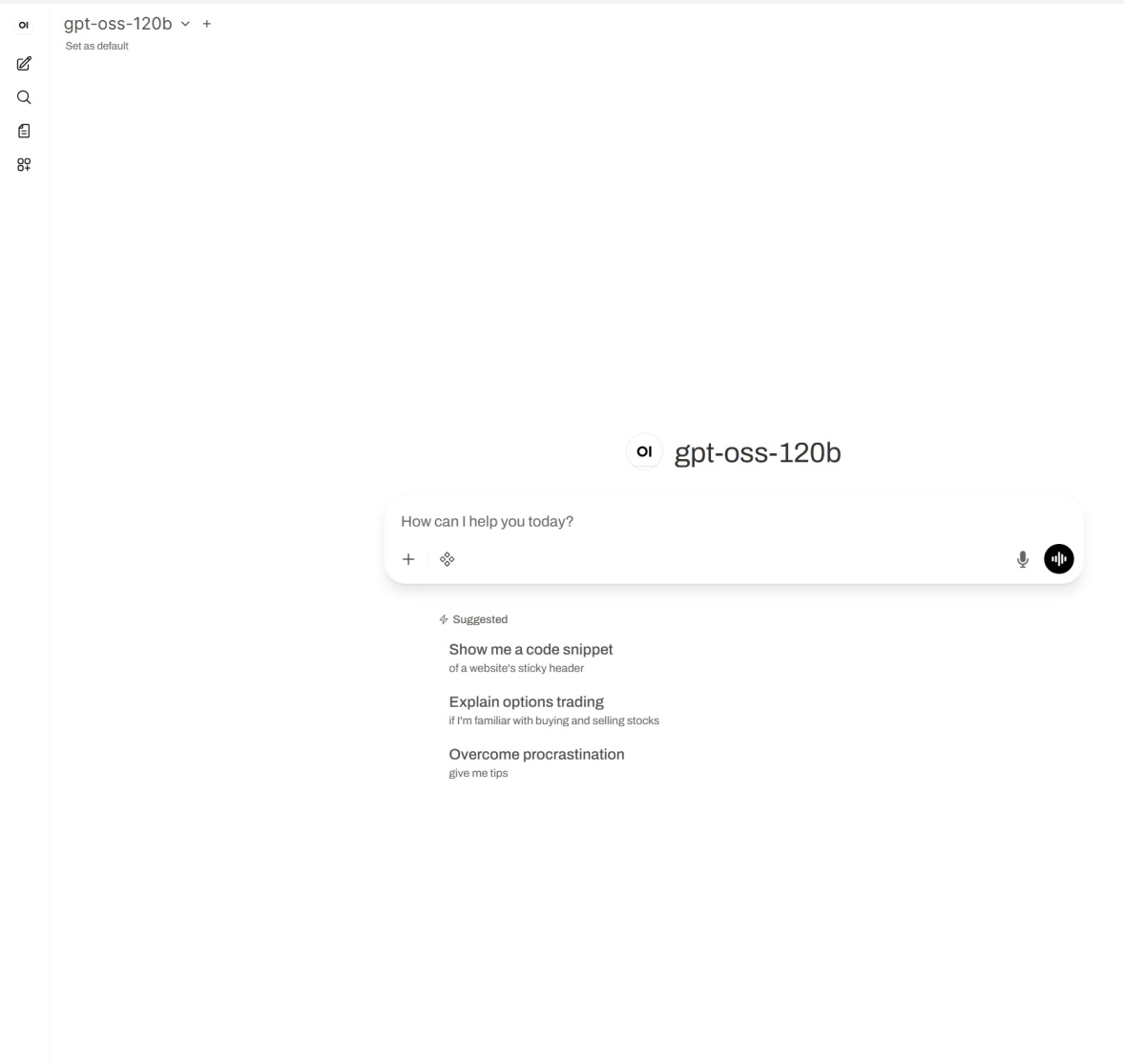
What You Did with Commands

- Learned how the workflow operates behind the scenes

Add a User Interface

- Turn the workflow into an application that anyone can use

OpenWebUI Screen



Next Steps: MCP and A2A

There are ways to connect the AI you built to other tools and AI systems.

MCP

A mechanism for connecting AI to external tools and data

For example, frontier models use this to search the internet for what they don't know.

A2A

A mechanism that allows AI systems to work together

AI systems with different roles divide the work and complete a single task

For today, just remember these names.

Summary

Great work. Here is what you can now do after this class.



Logged in to SAKURA AI Engine and selected a model



Issued an Account Token and called AI using curl



Registered a document and built a RAG chatbot



Explained the difference between results with and without RAG

Next

Take a break, then continue to Module 2.